NATIONAL BANK OF ROMANIA

Quantum Computing, Deep Learning and Financial Stability in Good Times

Alexie Alupoaiei Leonardo Badea Florian Neagu

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Introduction

- In recent years, we have witnessed an exponential increase in the use of modern tools in finance, such as Artificial Intelligence (AI), Machine Learning (ML) and, more recently, Quantum Computing

- On the lending side, these tools are used to automate loan approval decisions and assess the borrowers' ability to repay the contracted loans

- It is generally agreed that these tools have a greater predictive capacity for defaults than traditional quantitative tools

Motivation

- **First Goal**:

  - In this paper, we design how the intensive use of modern technologies for credit risk and loan pricing will affect financial stability

  - Here we focus exclusively on the pricing channel during favorable macroeconomic periods

  - Our investigation is a normative one

Motivation

- **Secondary Goals**:

  - To understand how this emergent tool, **quantum computing,** can be used in practical credit risk assessment

  - To explore how the quantum computing can be integrated efficiently within deep learning neural networks models

  - To navigate new opportunities of using a micro-macro approach for financial stability purposes

General Background

- Why Deep Learning? Because of its complexity and discipline

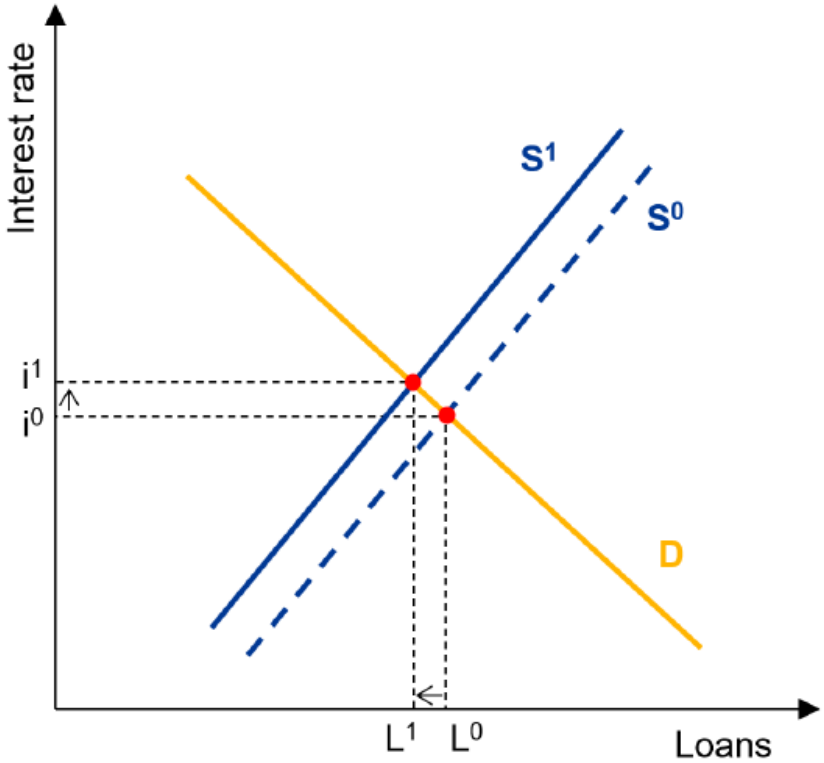
- Why Quantum Computing? Because of its (theoretical) high computation power

- Why Financial Stability in Good times? Because i) micro credit risk assessments are important for supervision and financial stability (Alonso and Carbó, 2021) and ii) financial cycles are asymmetric

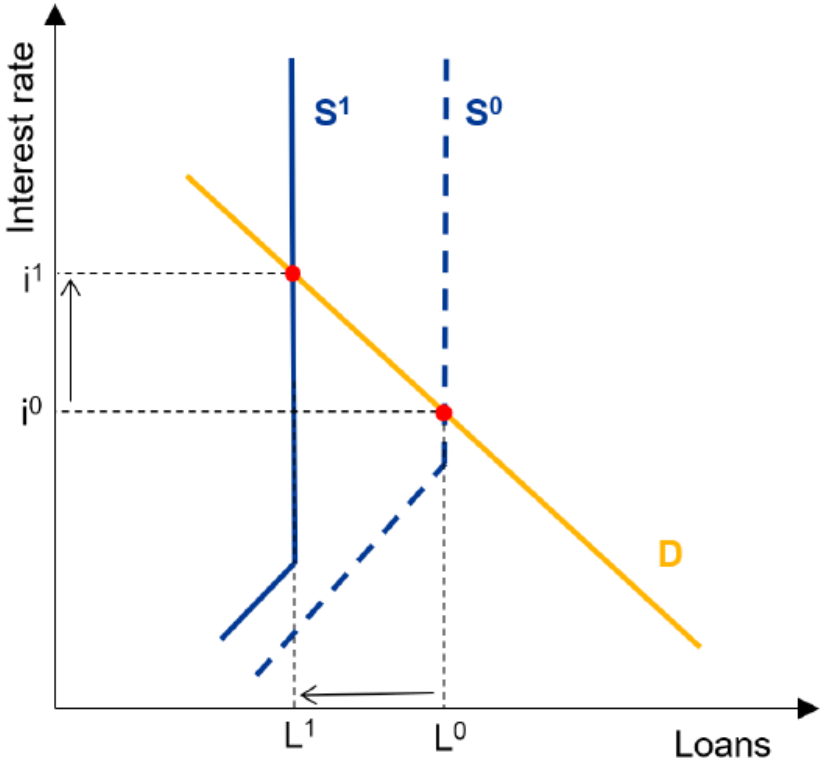
Effects of capital requirements over the business and financial cycles

- Stylised state-dependent transmission channels of capital requirements

(a) "Pricing channel": normal times



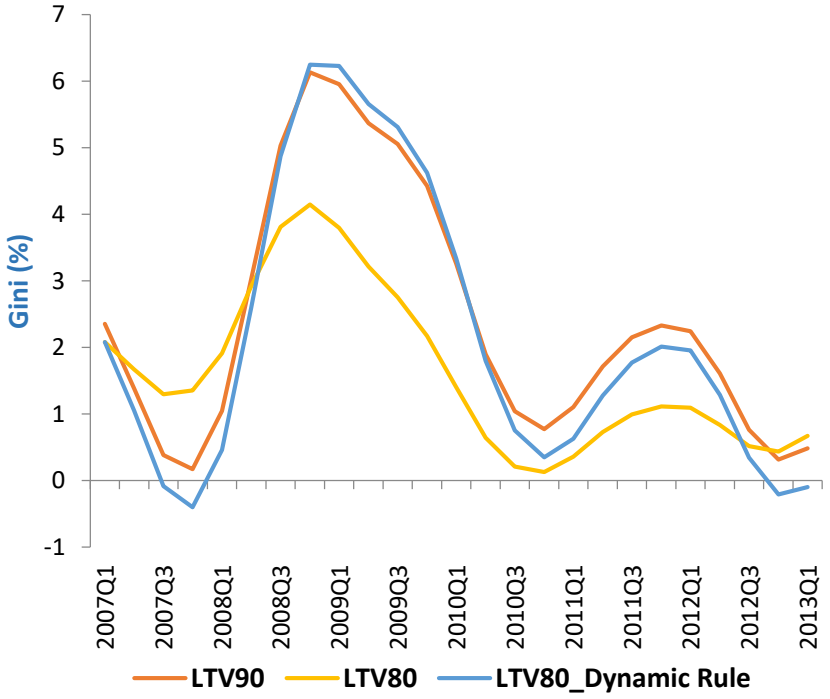
(b) "Quantity channel": crisis times

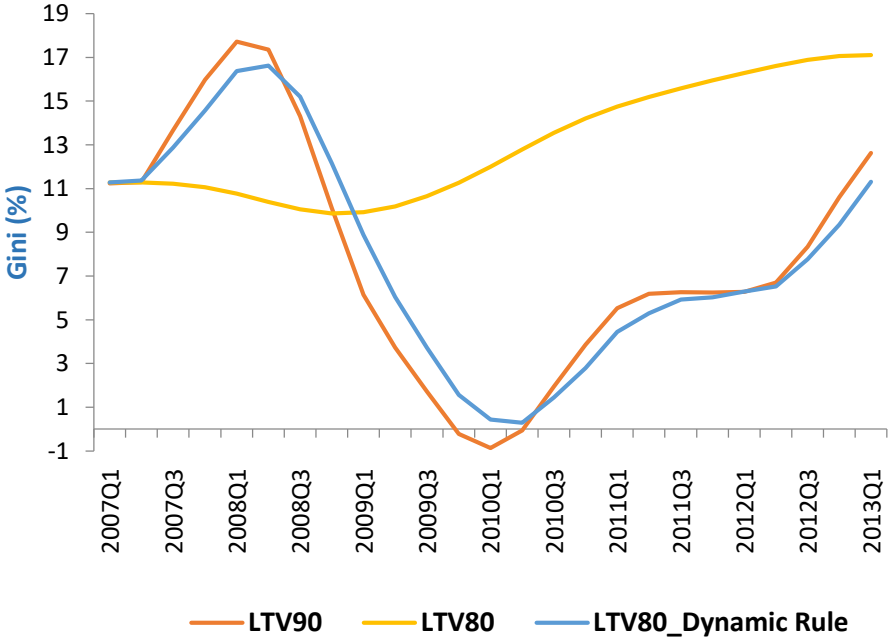


Source: Lang and Menno (2023)

Effects of borrower-based measures over the business and financial cycles

- Approximated (DSGE based) evolution of consumption and wealth inequality across GFC for different LTV policies





Source: Alupoaiei, Ichim, Neagu and Radu (2025)

Basic concepts about Quantum Computing

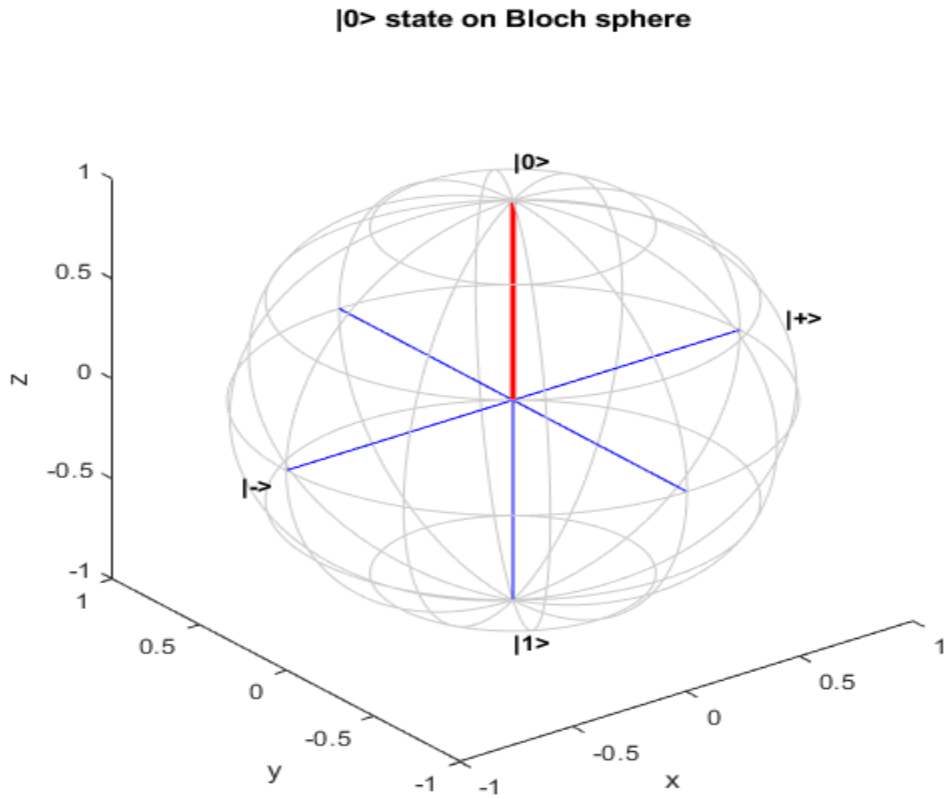
- Qubit – The basic unit of quantum computing

- Superposition – According to Erwin Schrodinger, place a cat with a deadly radioactive element in a sealed box and, in a quantum world, the cat is both dead and alive before the box is opened

- Entanglement – According to Abert Einstein, it's a "spooky action at a distance."

Basic concepts about Quantum Computing

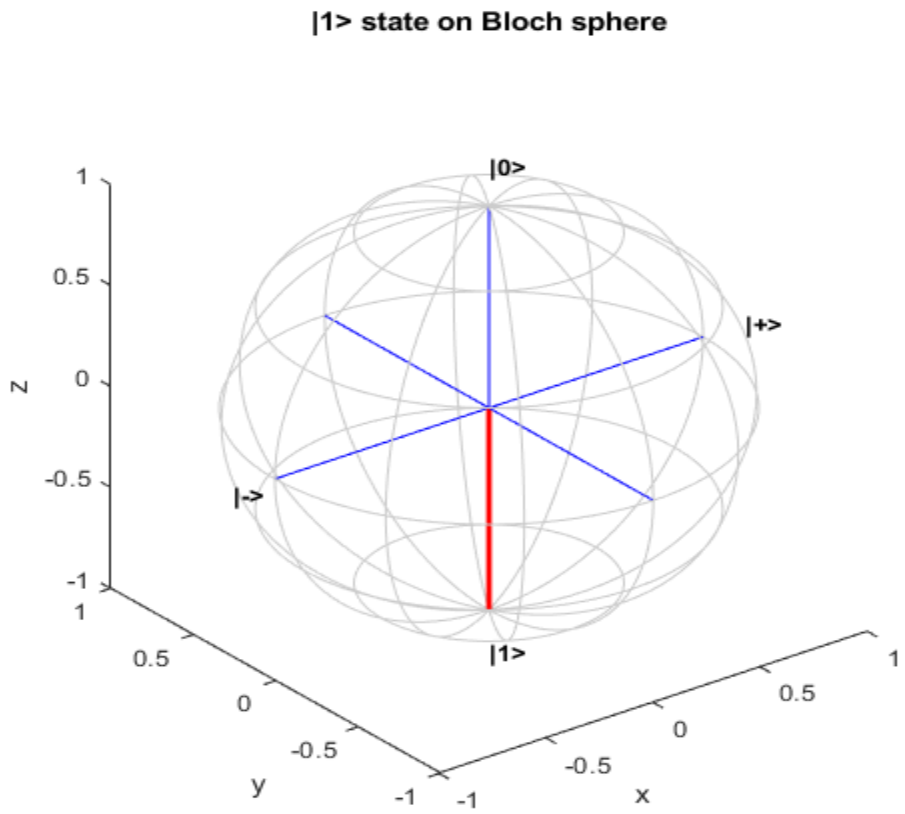
- Example of superpositions (grey lines) and |0⟩ being a basis state on the Bloch sphere



Source: authors' contribution

Basic Concepts about Quantum Computing

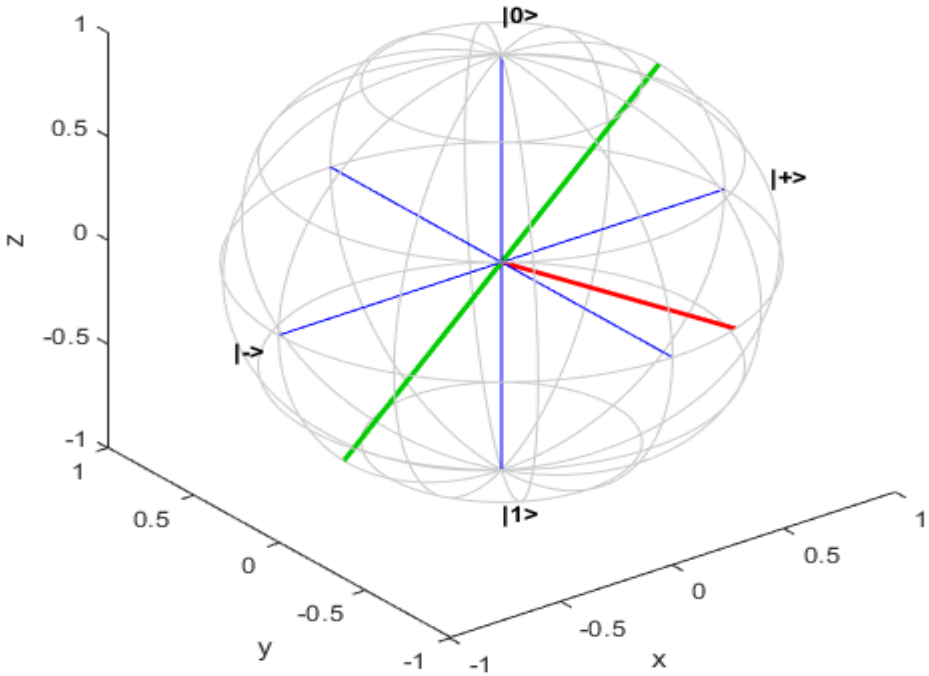
- Example of superpositions (grey lines) and |1⟩ being a basis state on the Bloch sphere



Source: authors' contribution

Basic concepts about Quantum Computing

- Superpositions of $|0\rangle$ and $|1\rangle$ with Hadamard gate transformations



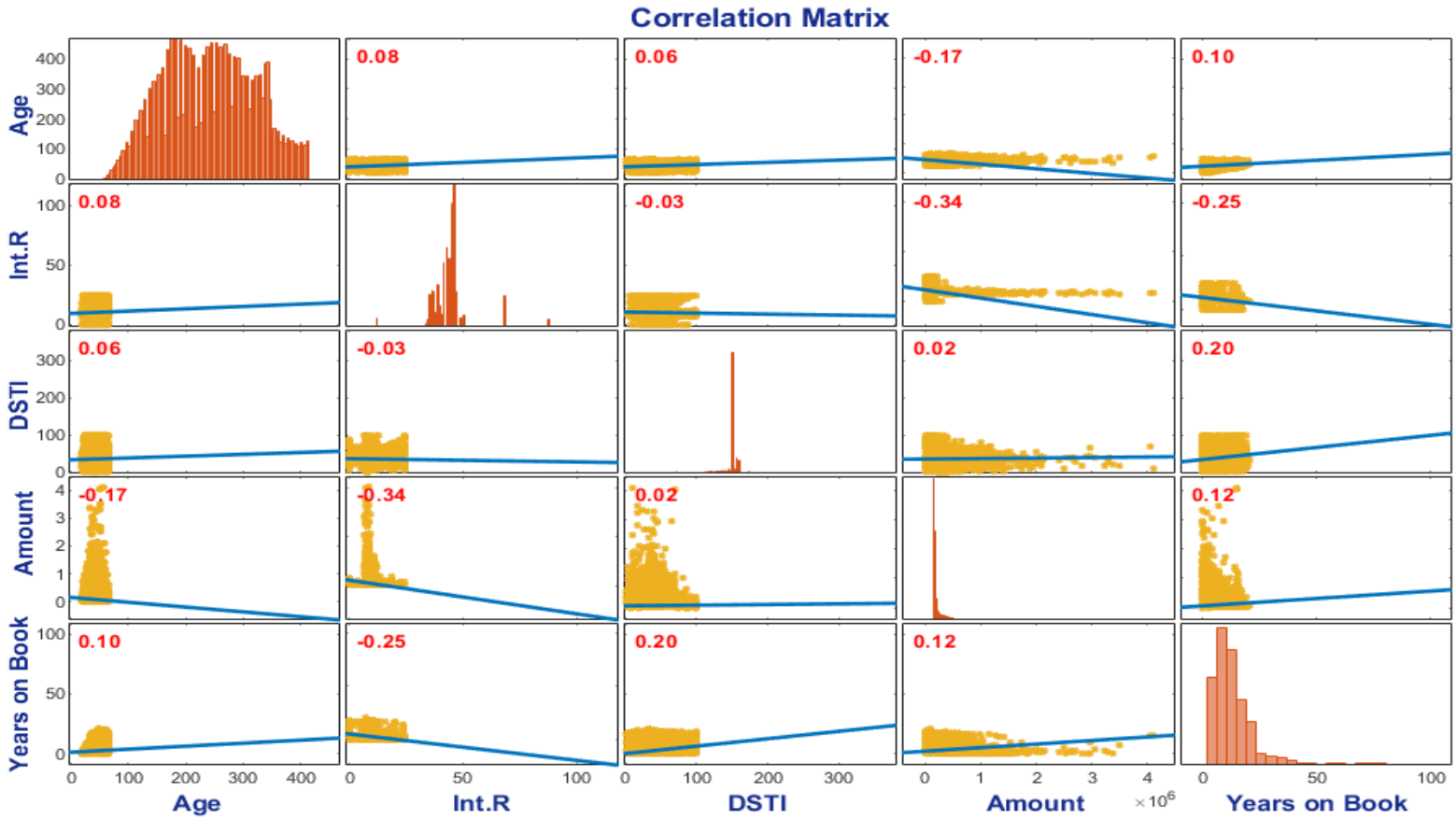
Source: authors' contribution

Sample used for investigation

- **Source**: Romanian Credit Registry
- **Number of loans**: 642,005
- **Types**: Consumer loans
- **Period under investigation**: June 2023 – June 2024
- **Default rate**: 2.35 %
- **Partition**: 80 % for training and 20 % for testing
- **Variables selected for estimation:** 'Age', ‘Interest Rate', 'DSTI', 'Loan Amount', 'Years on Book’
- **DSTI transformed into a score as: 1** for DSTI <= 20%, **2** for 20% < DSTI <= 40%, **3** for 40% < DSTI <= 60% and **4** for DSTI > 40%

Correlation matrix of the variables entering the model

- Selected do not show a strong correlation between them that could affect the statistical estimation of the PD model



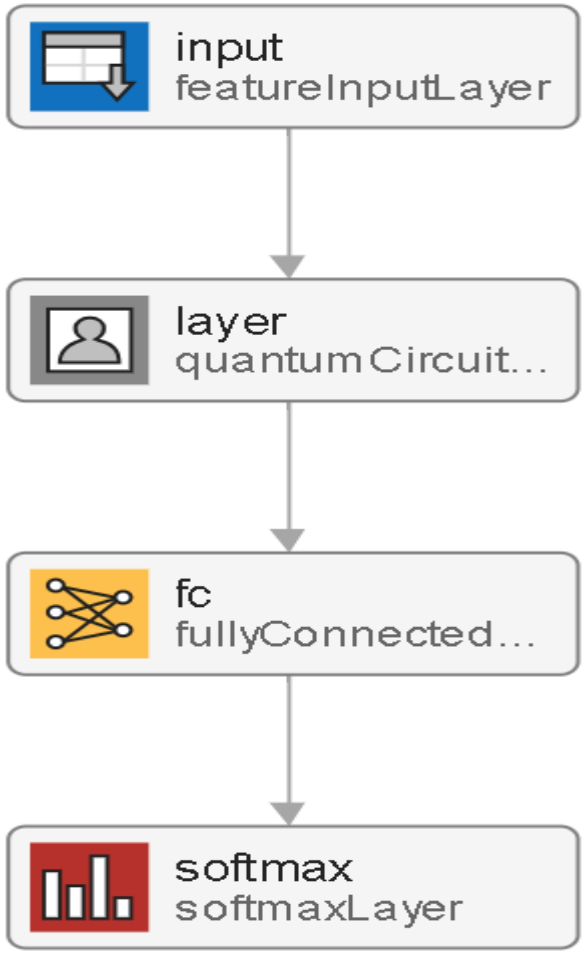
Source: authors' contribution

Merging Quantum Computing and Machine Learning

- Define the input feature vector
- Apply L2 (Euclidean) normalization
- Encode the data into a quantum state (amplitude encoding)
- Apply a parameterized quantum circuit (variational layer)
- Measure the quantum state (observable evaluation)
- Convert measurement to probability
- Perform binary classification
- Train the model (hybrid optimization)

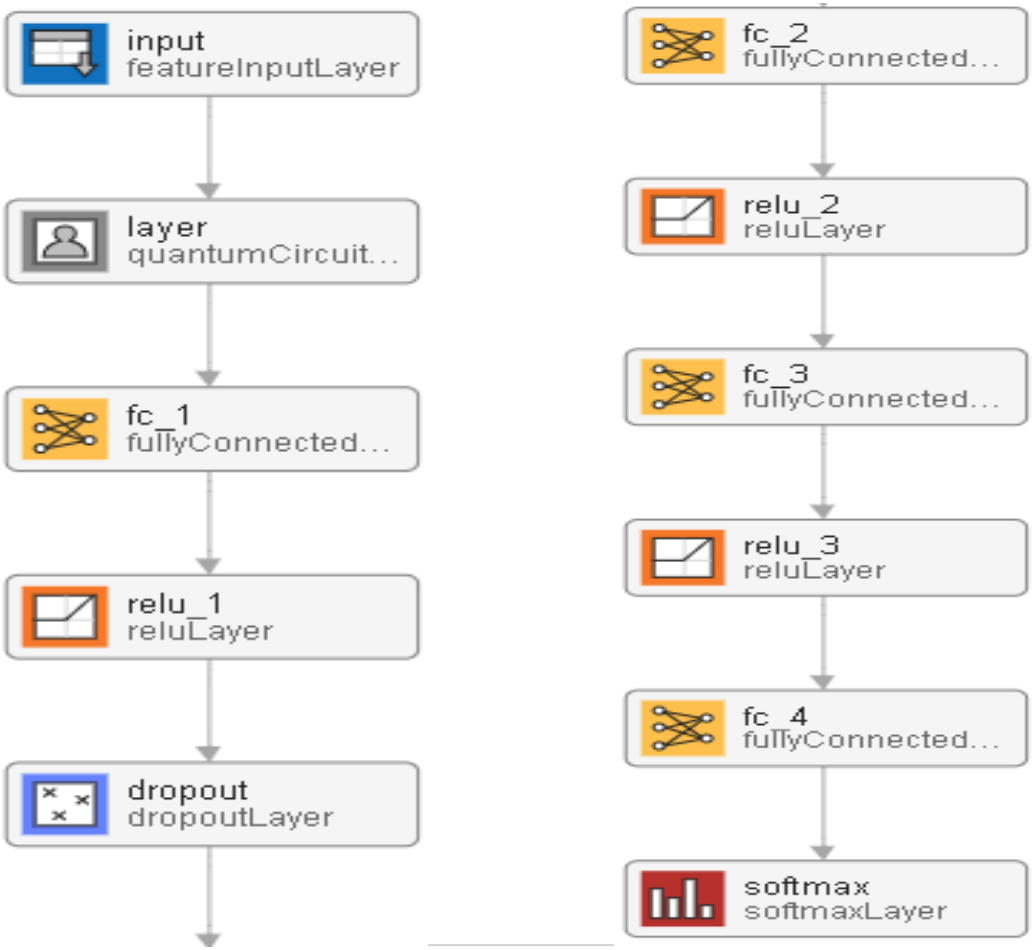
Quantum Computing and Deep Learning

- Design of a simple Quantum Neural Network model with 5 qubits



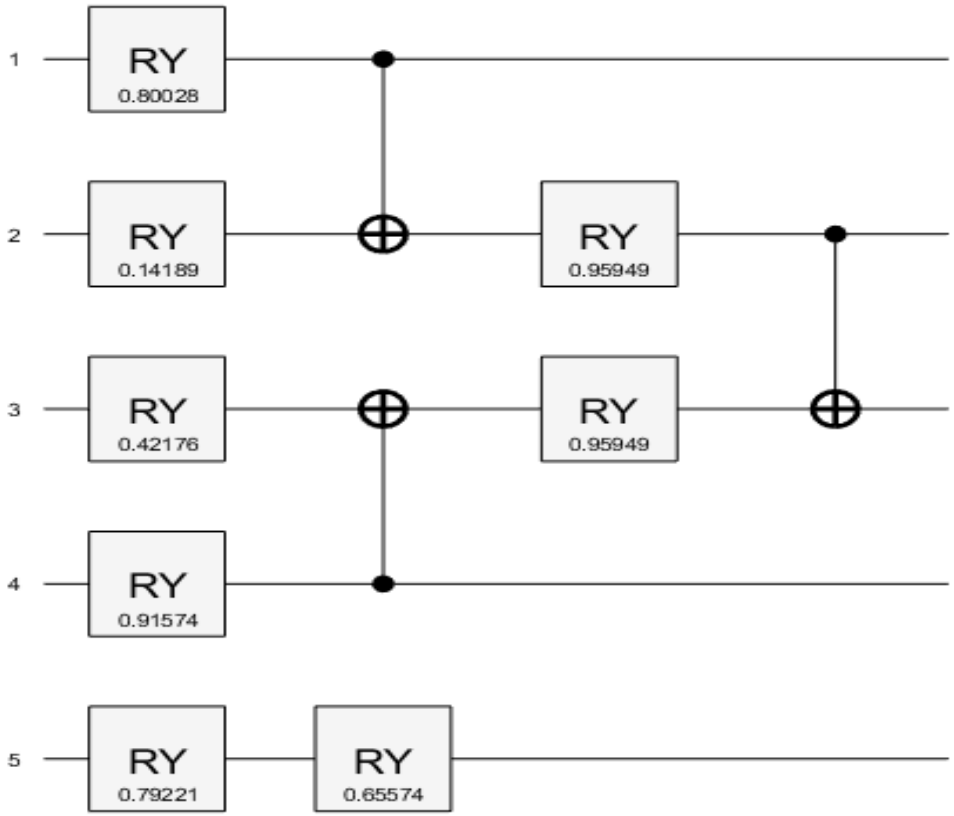
Quantum Deep Learning Network Architecture

- Following the deep learning architecture used by Alupoaiei, Neagu and Negrea (2025), we attached a quantum layer to encode the input data into quantum states



Quantum Gates for Credit Classification

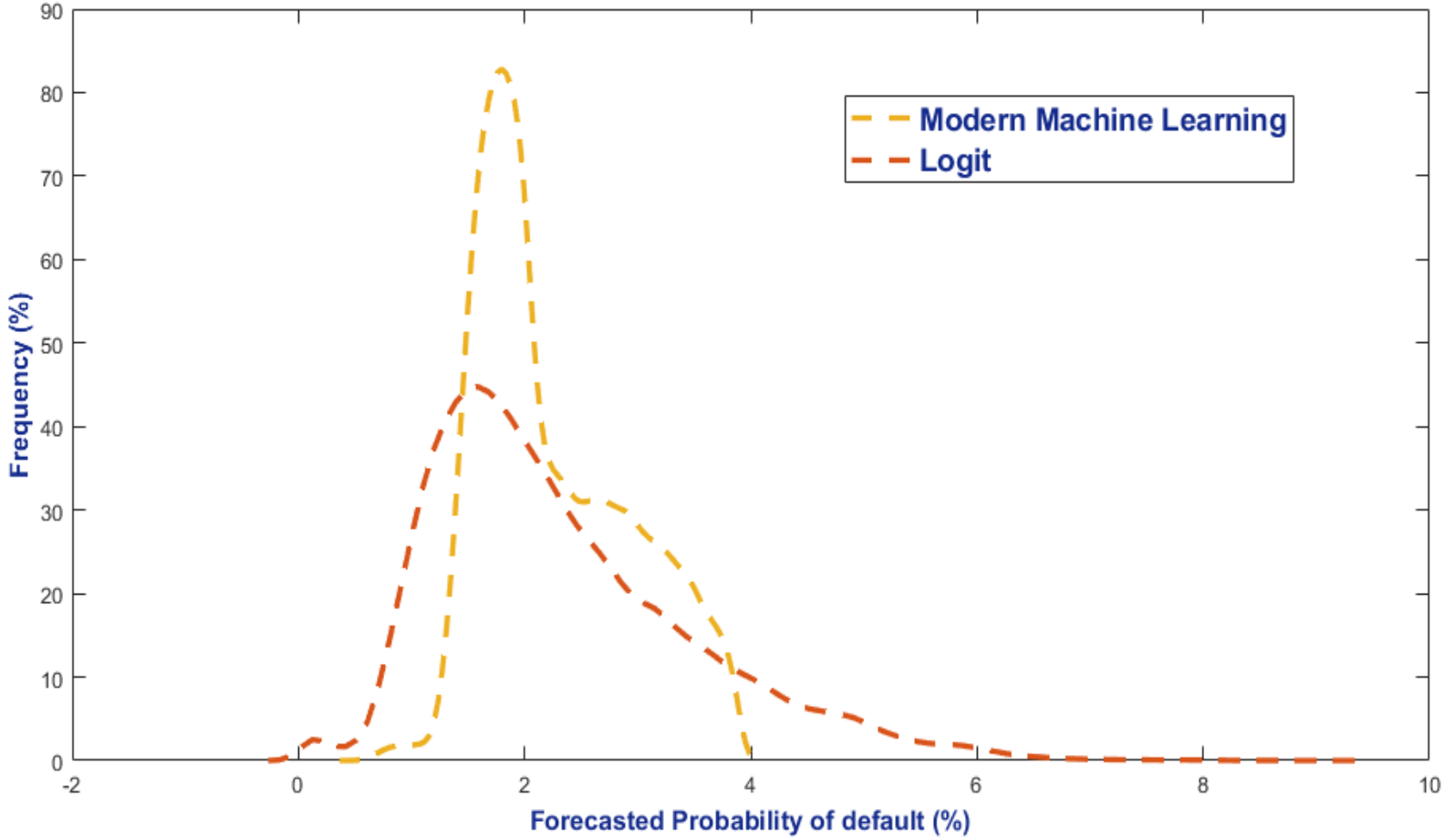
- We define a combination of RY and CNOT gates for 5 qubits and 7 learnable parameters



Source: authors' contribution

Modern Technologies and Financial Stability

- Estimated densities for the forecasted PDs



Source: authors' contribution

Modern Technologies and Financial Stability

- We calibrate the partial equilibrium model of Suárez and Repullo (2003) to investigate the effects of estimated PDs on loan pricing

	Lending Rates				
Quantiles	5 %	10 %	50 % (median)	90 %	95 %
Modern Machine Learning	2.11	2.23	3.24	5.3	5.42
Logit	0.96	1.25	3.26	7.19	7.71

Source: authors' contribution

Merging Quantum Computing and Neural Networks for credit risk assessments – some considerations

- Even though QNNs can theoretically separate non-linearly inseparable classes, their performance strongly depends on how they are parameterized and interpreted

- For practical micro-level credit assessment, a significant amount of computational power is required for QNNs to be operational

- For macro-level credit assessment, applying techniques such as SMOTE or bootstrap-based methods can be effective for dedicated analyses

Conclusions

- Some potential effects associated with the widespread use of modern technologies for credit classification are related to:
  - Effects of lowering loan interest rates during favorable economic periods
  - Effects of lowering some benchmark rates, such as R*
  - Effects on the transmission channels of monetary policy
  - Shaping some perspectives for macroprudential policymakers regarding financial stability

Thank you for your attention!